Hardware Simulations of Open Quantum Systems by First-Order Hamiltonian Dilation

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Abstract

This note gives a workflow for simulating Lindblad dynamics on IBM-style superconducting hardware using the first-order Hamiltonian dilation method of Ding–Li–Lin, together with mid-circuit measurement/reset and optional step-size/zero-noise extrapolation. The emphasis is not asymptotic complexity, but rather model problems that can be implemented and benchmarked on current devices: a two-qubit damped exchange model, a boundary-driven spin chain with measurable nonequilibrium current, and a quantum-detailed-balance thermalization model. The central practical feature is that one physical Lindblad step of size Δt is implemented by a coherent unitary whose dissipative coupling angles scale as $\sqrt{\Delta t}$, followed by tracing out or resetting the ancilla.

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1 Conventions and target hardware workflow

We use the computational convention

$$Z\,|0\rangle = +\,|0\rangle\,, \qquad Z\,|1\rangle = -\,|1\rangle\,, \tag{1}$$

so that

$$\sigma^- = |0\rangle\langle 1| = \frac{X+iY}{2}, \qquad \sigma^+ = |1\rangle\langle 0| = \frac{X-iY}{2}. \tag{2}$$

Thus σ^- is the amplitude-damping/lowering operator from $|1\rangle$ to $|0\rangle$.

The hardware loop for one trajectory-free Lindblad simulation is:

Step 1. prepare the system state ρ_0;

Step 2. prepare the ancilla register in $|0\rangle_a$;

Step 3. apply a dilation unitary $U_{\Delta t} = \exp\Big(-i\sqrt{\Delta t}\,\widetilde{H}_{\Delta t}\Big)$;

Step 4. measure and reset the ancilla, or equivalently trace it out;

Step 5. repeat for $N = T/\Delta t$ steps;

Step 6. measure system observables in appropriate Pauli bases.

No postselection is used: all ancilla outcomes are retained. The ancilla measurement record can be stored as a diagnostic of effective jump statistics, but the Lindblad estimator is the unconditional average over all outcomes.

2 Lindblad equation and first-order Hamiltonian dilation

The Markovian open-system dynamics is

$$\frac{d\rho}{dt} = \mathcal{L}(\rho) = -i[H,\rho] + \sum_{j=1}^{J}\left(V_j \rho V_j^\dagger - \frac{1}{2}\{V_j^\dagger V_j, \rho\}\right), \tag{3}$$

where $H = H^\dagger$ is the system Hamiltonian and V_j are jump operators. The first-order dilation used here follows the Ding–Li–Lin construction [6] and is closely related in spirit to earlier quantum algorithms for Lindblad simulation [4, 13].

For one time step Δt, define an ancilla space with basis states

$$|0\rangle_a, |1\rangle_a, \ldots, |J\rangle_a. \tag{4}$$

The first-order dilated Hamiltonian is

$$\widetilde{H}_{\Delta t} = \begin{bmatrix} \sqrt{\Delta t}\, H & V_1^\dagger & V_2^\dagger & \cdots & V_J^\dagger \\ V_1 & 0 & 0 & \cdots & 0 \\ V_2 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ V_J & 0 & 0 & \cdots & 0 \end{bmatrix}. \tag{5}$$

Equivalently,

$$\widetilde{H}_{\Delta t} = \sqrt{\Delta t}\, |0\rangle\langle 0|_a \otimes H + \sum_{j=1}^{J}\left(|j\rangle\langle 0|_a \otimes V_j + |0\rangle\langle j|_a \otimes V_j^\dagger\right). \tag{6}$$

The corresponding channel is

$$\Phi_{\Delta t}(\rho) = \mathrm{Tr}_a\left[U_{\Delta t}(|0\rangle\langle 0|_a \otimes \rho) U_{\Delta t}^\dagger\right], \qquad U_{\Delta t} = \exp\left(-i\sqrt{\Delta t}\,\widetilde{H}_{\Delta t}\right). \tag{7}$$

It satisfies

$$\Phi_{\Delta t}(\rho) = \rho + \Delta t\, \mathcal{L}(\rho) + O(\Delta t^2), \tag{8}$$

so $N = T/\Delta t$ repetitions give a first-order approximation over time T.

2.1 The important $\sqrt{\Delta t}$ scale

The exponent can be written as

$$\sqrt{\Delta t}\,\widetilde{H}_{\Delta t} = \underbrace{\Delta t\, |0\rangle\langle 0|_a \otimes H}_{\text{coherent system term}} + \underbrace{\sqrt{\Delta t}\sum_{j=1}^{J}\left(|j\rangle\langle 0|_a \otimes V_j + \text{h.c.}\right)}_{\text{dissipative dilation terms}}. \tag{9}$$

This is the central practical point. The physical Lindblad time step is Δt, but the jump-coupling rotations have angles proportional to $\sqrt{\Delta t}$. Therefore reducing Δt has two competing effects:

- smaller algorithmic error, $O(T\Delta t)$ for the first-order method;
- more repetitions $N = T/\Delta t$, hence more reset/readout/gate noise;
- only a $\sqrt{\Delta t}$ reduction in the dissipative rotation angles.

This makes the problem ideal for step-size extrapolation and hardware zero-noise extrapolation [14, 20, 11].

2.2 Density-operator initialization

On hardware, the density operator is not loaded as a matrix. We use one of the following procedures.

Pure initial state. Prepare $\rho_0 = |\psi_0\rangle\langle\psi_0|$ by an ordinary state-preparation circuit. This is the default for the two-qubit damping and current-buildup benchmarks.

Classical mixture. If

$$\rho_0 = \sum_\ell p_\ell |\psi_\ell\rangle\langle\psi_\ell|, \tag{10}$$

run separate jobs for the pure states $|\psi_\ell\rangle$ and combine measured expectations classically:

$$\langle O\rangle_{\rho_0}(t) = \sum_\ell p_\ell \langle O\rangle_{\psi_\ell}(t). \tag{11}$$

For diagonal Gibbs or infinite-temperature initial states, this is particularly efficient because $|\psi_\ell\rangle$ are computational-basis states.

Purification. A purification of ρ_0 can be used, but it costs extra qubits. For current IBM hardware, use classical mixture sampling unless a coherent reference system is essential.

2.3 Generic first-order Trotterization

Let

$$K_{\Delta t} := \sqrt{\Delta t}\,\widetilde{H}_{\Delta t} = K_H + \sum_{j=1}^{J} K_j, \tag{12}$$

where

$$K_H = \Delta t\ |0\rangle\langle 0|_a \otimes H, \qquad K_j = \sqrt{\Delta t}\left(|j\rangle\langle 0|_a \otimes V_j + |0\rangle\langle j|_a \otimes V_j^\dagger\right). \tag{13}$$

A first-order product formula is

$$e^{-iK_{\Delta t}} = e^{-iK_H}\prod_{j=1}^{J} e^{-iK_j} + O(\|K\|^2), \tag{14}$$

while a symmetric formula is

$$e^{-iK_{\Delta t}} = e^{-iK_H/2}\left(\prod_{j=1}^{J} e^{-iK_j/2}\right)\left(\prod_{j=J}^{1} e^{-iK_j/2}\right)e^{-iK_H/2} + O(\|K\|^3). \tag{15}$$

Because the Lindblad method itself is first order, eq. (15) is mainly useful to reduce circuit-synthesis error so that the observed time-step error is dominated by the intended Lindblad discretization, not by the Hamiltonian-simulation subroutine.

3 Model problem I: two-qubit damped exchange model

3.1 System model

The first hardware target should be a two-qubit exchange Hamiltonian with local damping:

$$H_2 = \frac{J}{2}(X_1X_2 + Y_1Y_2) + \frac{h}{2}(Z_1 + Z_2), \tag{16}$$

with jump operators

$$V_1 = \sqrt{\gamma}\,\sigma_1^-, \qquad V_2 = \sqrt{\gamma}\,\sigma_2^-. \tag{17}$$

This is a minimal model with coherent excitation exchange and irreversible loss. It is small enough for exact classical reference simulation, and it exposes the essential hardware difficulty: repeated ancilla reset after dilation.

Useful starting parameters are

$$J = 1, \qquad h = 0, \qquad \gamma \in \{0.1, 0.2, 0.5\}, \qquad \Delta t \in \{0.05, 0.1, 0.2\}. \tag{18}$$

Use dimensionless units with $\hbar = 1$. The expected relaxation time is $O(1/\gamma)$, so use

$$T \approx 2/\gamma \text{ to } 5/\gamma \tag{19}$$

for steady decay tests, but use shorter times $T \lesssim 1/\gamma$ for high-fidelity transient-current tests.

3.2 Ancilla encoding and explicit dilated Hamiltonian

There are $J = 2$ jumps, so the minimal ancilla has dimension 3. On qubit hardware, encode

$$|0\rangle_a \equiv |00\rangle_a\,, \qquad |1\rangle_a \equiv |01\rangle_a\,, \qquad |2\rangle_a \equiv |10\rangle_a\,, \tag{20}$$

leaving $|11\rangle_a$ unused. In the active qutrit subspace,

$$\widetilde{H}^{(2)}_{\Delta t} = \begin{bmatrix} \sqrt{\Delta t}\,H_2 & V_1^\dagger & V_2^\dagger \\ V_1 & 0 & 0 \\ V_2 & 0 & 0 \end{bmatrix}. \tag{21}$$

Equivalently, defining $E_{mn} = |m\rangle\langle n|_a$,

$$K^{(2)}_{\Delta t} := \sqrt{\Delta t}\,\widetilde{H}^{(2)}_{\Delta t} = \Delta t\,E_{00}\otimes H_2 + \sqrt{\Delta t}\sum_{q=1}^{2}\left(E_{q0}\otimes V_q + E_{0q}\otimes V_q^\dagger\right). \tag{22}$$

For amplitude damping in our convention,

$$V_q = \frac{\sqrt{\gamma}}{2}(X_q + iY_q), \tag{23}$$

so if

$$A^x_{q0} = E_{q0} + E_{0q}, \qquad A^y_{q0} = -i(E_{q0} - E_{0q}), \tag{24}$$

then

$$E_{q0}\otimes V_q + E_{0q}\otimes V_q^\dagger = \frac{\sqrt{\gamma}}{2}\left(A^x_{q0}\otimes X_q - A^y_{q0}\otimes Y_q\right). \tag{25}$$

The two terms in eq. (25) commute within the active two-level subspace $\mathrm{span}\{|0\rangle_a\,, |q\rangle_a\}$, so each local damping block is an exchange-like $XX + YY$ rotation between the ancilla transition and the system qubit. This is a useful circuit optimization.

3.3 Trotter simplifications for the two-qubit model

For eq. (16), the terms X_1X_2, Y_1Y_2, and $Z_1 + Z_2$ mutually commute when the field is uniform. Therefore e^{-iK_H} can be compiled exactly as controlled Pauli rotations under the control projector E_{00}:

$$e^{-i\Delta t E_{00}\otimes H_2} = e^{-i\frac{J\Delta t}{2}E_{00}\otimes X_1X_2}e^{-i\frac{J\Delta t}{2}E_{00}\otimes Y_1Y_2}e^{-i\frac{h\Delta t}{2}E_{00}\otimes Z_1}e^{-i\frac{h\Delta t}{2}E_{00}\otimes Z_2}. \tag{26}$$

The jump terms K_1 and K_2 do not generally commute because they share the same ancilla state $|0\rangle_a$. Use the symmetric ordering

$$U_{\Delta t}^{\text{trotter}} = e^{-iK_H/2}e^{-iK_1/2}e^{-iK_2}e^{-iK_1/2}e^{-iK_H/2} \tag{27}$$

for the first hardware benchmark. If the compiled depth is too high, compare against the cheaper ordering $e^{-iK_H}e^{-iK_1}e^{-iK_2}$ and use exact classical simulation to separate Trotter error from device noise.

3.4 Initial states and expected behavior

Use the following initial states:

$$\rho_0^{(1)} = |10\rangle\langle 10|, \qquad \text{localized excitation,} \tag{28}$$

$$\rho_0^{(2)} = |01\rangle\langle 01|, \qquad \text{opposite localized excitation,} \tag{29}$$

$$\rho_0^{(+)} = |\psi_+\rangle\langle\psi_+|, \quad |\psi_+\rangle = \frac{|10\rangle + |01\rangle}{\sqrt{2}}, \qquad \text{one-excitation eigenstate for } h = 0. \tag{30}$$

For $\gamma > 0$, all initial states should relax toward $|00\rangle\langle 00|$ in this convention. For $|10\rangle$ or $|01\rangle$, one should see coherent exchange oscillations damped by the loss. For $|\psi_+\rangle$, the exchange part is stationary when $h = 0$, so the signal is mostly damping.

3.5 Measurements

Measure:

$$n_q(t) = \frac{1 - \langle Z_q(t)\rangle}{2}, \qquad q = 1, 2, \tag{31}$$

$$C_{XY}(t) = \langle X_1X_2 + Y_1Y_2\rangle, \tag{32}$$

$$j_{12}(t) = J\langle X_1Y_2 - Y_1X_2\rangle. \tag{33}$$

Here n_q is the excitation population on qubit q. The current j_{12} follows from the continuity equation for the exchange Hamiltonian in eq. (16). On hardware, j_{12} requires two Pauli-basis measurements: X_1Y_2 and Y_1X_2.

Expected signatures:

- $n_1 + n_2$ decays approximately exponentially on a time scale $1/\gamma$;
- $n_1 - n_2$ oscillates with frequency set by J and decays due to damping;
- $j_{12}(t)$ is transient for localized initial states and tends to zero at long times;
- step-size extrapolation in Δt should improve agreement with exact Lindblad evolution until hardware noise dominates.

3.6 Circuit Construction Possiblilities

The following circuit constructions are possible for Model I, more details later : i) Unitary Circuits : Simplified for both symmetric and first order but quantum control operations due to projectors could lead to large depth ii) Dynamic Circuits : We need one quantum control and 2 classical controls for symmetric case but for first order we could just reduce it to one classical control iii) Qubit Reuse : For the first order, it could further be simplified by just using 1 ancilla rather than j ancillas using dynamic circuit approach.

4 Model problem II: boundary-driven XXZ chain and steady current

4.1 Hamiltonian and boundary jumps

The ambitious but still realistic target is a boundary-driven XXZ chain on $n = 2$ or $n = 3$ system qubits:

$$H_{\mathrm{XXZ}} = \sum_{\ell=1}^{n-1} \left[\frac{J}{2}(X_\ell X_{\ell+1} + Y_\ell Y_{\ell+1}) + \frac{\Delta}{4} Z_\ell Z_{\ell+1} \right]. \tag{34}$$

The Z-magnetization current across bond $\ell, \ell+1$ is

$$j_\ell = J\left(X_\ell Y_{\ell+1} - Y_\ell X_{\ell+1}\right), \tag{35}$$

with the above normalization.

The boundary reservoirs are represented by four jump operators:

$$V_{L,+} = \sqrt{\frac{\Gamma}{2}(1+\mu)}\,\sigma_1^+, \qquad V_{L,-} = \sqrt{\frac{\Gamma}{2}(1-\mu)}\,\sigma_1^-, \tag{36}$$

$$V_{R,+} = \sqrt{\frac{\Gamma}{2}(1-\mu)}\,\sigma_n^+, \qquad V_{R,-} = \sqrt{\frac{\Gamma}{2}(1+\mu)}\,\sigma_n^-. \tag{37}$$

For an isolated boundary spin, the left bath favors magnetization $-\mu$ while the right bath favors magnetization $+\mu$. Thus μ is the driving bias and Γ is the system–bath coupling. This model is standard in studies of nonequilibrium steady states of boundary-driven spin chains [15, 16, 8, 12].

Recommended initial parameter choices:

$$n = 2, 3, \qquad J = 1, \qquad \Delta \in \{0, 0.5, 1.5\}, \qquad \Gamma \in \{0.1, 0.2, 0.5\}, \qquad \mu \in \{0.1, 0.5\}. \tag{38}$$

Use $\mu = 0.1$ for linear-response comparisons and $\mu = 0.5$ for a larger hardware signal.

4.2 Dilated Hamiltonian

There are $J_{\mathrm{jump}} = 4$ jumps, so the ancilla dimension is 5. Use a 3-qubit ancilla register and encode active states

$$|0\rangle_a, |L+\rangle_a, |L-\rangle_a, |R+\rangle_a, |R-\rangle_a, \tag{39}$$

leaving the other computational states unused. The first-order dilation is

$$\widetilde{H}_{\Delta t}^{\mathrm{bdry}} = \sqrt{\Delta t}\,|0\rangle\langle 0|_a \otimes H_{\mathrm{XXZ}} + \sum_{\alpha \in \{L+, L-, R+, R-\}} \left(|\alpha\rangle\langle 0|_a \otimes V_\alpha + |0\rangle\langle \alpha|_a \otimes V_\alpha^\dagger \right). \tag{40}$$

The unitary per Lindblad step is

$$U_{\Delta t}^{\mathrm{bdry}} = \exp\Big(-i\sqrt{\Delta t}\,\widetilde{H}_{\Delta t}^{\mathrm{bdry}}\Big). \tag{41}$$

For compilation, define

$$K_{\Delta t}^{\mathrm{bdry}} = \sqrt{\Delta t}\,\widetilde{H}_{\Delta t}^{\mathrm{bdry}} = \Delta t\;|0\rangle\langle 0|_a \otimes H_{\mathrm{XXZ}} + \sqrt{\Delta t}\sum_{\alpha}\left(|\alpha\rangle\,\langle 0|_a \otimes V_\alpha + \mathrm{h.c.}\right). \tag{42}$$

Again, the Hamiltonian part has $O(\Delta t)$ angles, while each boundary jump coupling has $O(\sqrt{\Delta t\,\Gamma})$ angles.

4.3 Trotter strategy

For $n = 2$, all Hamiltonian terms in eq. (34) act on the same bond and the three Pauli strings XX, YY, and ZZ commute, so the controlled Hamiltonian block can be synthesized exactly as controlled Pauli rotations.

For $n \geq 3$, use an even–odd bond decomposition:

$$H_{\mathrm{XXZ}} = H_{\mathrm{even}} + H_{\mathrm{odd}}, \tag{43}$$

where each part is a sum of disjoint two-qubit bond Hamiltonians. A symmetric step is

$$e^{-iK_{\Delta t}^{\mathrm{bdry}}} \approx e^{-iK_H/2}\left(\prod_{\alpha} e^{-iK_\alpha/2}\right)\left(\prod_{\alpha}^{\mathrm{reverse}} e^{-iK_\alpha/2}\right) e^{-iK_H/2}, \tag{44}$$

where $K_H = \Delta t\,|0\rangle\langle 0|_a \otimes H_{\mathrm{XXZ}}$ and K_α is the αth boundary jump block. The order of the four boundary jumps should be fixed across all time steps; reversing the order in the second half gives a Strang-type cancellation of leading synthesis error.

Possible simplifications:

- for $n = 2$, compile the full active-subspace unitary exactly and compare with Pauli product formulas;
- for $\Delta = 0$, the XX model has fewer noncommuting Hamiltonian terms;
- for weak drive $\mu \ll 1$, collect data for both $+\mu$ and $-\mu$ and antisymmetrize the current to cancel even-in-μ biases;
- use Pauli grouping for $X_\ell X_{\ell+1}$ and $Y_\ell Y_{\ell+1}$ rotations, which commute on each bond.

4.4 Initial states, evolution time, and steady-state certification

Good initial states are:

$$\rho_0 = |00\cdots 0\rangle\,\langle 00\cdots 0|\,, \tag{45}$$

$$\rho_0 = 2^{-n}\mathbb{I}, \quad \text{implemented by uniform computational-basis sampling,} \tag{46}$$

$$\rho_0 = |10\cdots 0\rangle\,\langle 10\cdots 0|\,, \quad \text{for current-buildup tests.} \tag{47}$$

The steady-state time should be estimated classically for each n and parameter set by diagonalizing the small Liouvillian or by exact time integration. Let λ_{gap} denote the Liouvillian spectral gap, i.e. the smallest nonzero decay rate. A practical target is

$$T \gtrsim \frac{3}{\lambda_{\mathrm{gap}}} \quad \text{for a visible plateau,} \qquad T \gtrsim \frac{5}{\lambda_{\mathrm{gap}}} \quad \text{for quantitative steady-state estimates.} \tag{48}$$

For hardware, also run shorter transients because the current buildup may be more robust than the final long-time plateau.

Certify stationarity by checking:

$$|\langle Z_j(t+\Delta t)\rangle - \langle Z_j(t)\rangle| \ll 1, \tag{49}$$

$$|\langle j_\ell(t+\Delta t)\rangle - \langle j_\ell(t)\rangle| \ll 1, \tag{50}$$

$$\langle j_1\rangle \approx \langle j_2\rangle \approx \cdots \approx \langle j_{n-1}\rangle \quad \text{in the bulk for } n \geq 3. \tag{51}$$

4.5 What to measure and how to interpret it

Measure the magnetization profile and current:

$$m_j = \langle Z_j\rangle, \qquad \langle j_\ell\rangle = J\left(\langle X_\ell Y_{\ell+1}\rangle - \langle Y_\ell X_{\ell+1}\rangle\right). \tag{52}$$

Expected behavior:

- $m_1 < m_n$ for $\mu > 0$;
- the steady current changes sign when $\mu \mapsto -\mu$;
- for small μ, $\langle j\rangle$ should be approximately linear in μ;
- the current magnitude depends on Δ, with integrable/ballistic-like behavior near the XX regime and more suppressed transport for stronger anisotropy in small systems.

4.6 Relation to linear response and autocorrelations

In the weak-driving regime $\mu \ll 1$ and weak system–bath coupling $\Gamma \ll J$, the boundary-driven nonequilibrium steady current is a finite-size open-system proxy for a linear-response transport coefficient. In closed equilibrium systems, the Green–Kubo relation expresses transport coefficients in terms of equilibrium current autocorrelations,

$$\kappa_L(\beta) \sim \int_0^\infty C_{JJ}(t)\,dt, \qquad C_{JJ}(t) = \frac{1}{L}\langle J_{\mathrm{tot}}(t)J_{\mathrm{tot}}(0)\rangle_\beta, \tag{53}$$

where

$$J_{\mathrm{tot}} = \sum_{\ell=1}^{n-1} j_\ell. \tag{54}$$

The equality between a boundary-driven finite-size NESS current and the closed-system Green–Kubo coefficient is not automatic. It is a comparison to make in the linear-response, weak-coupling, finite-size regime. Recent studies of boundary-driven spin chains explicitly discuss this connection and its limitations [8, 12]. For the summer project, the practical goal is to measure the NESS current and compare its small-μ slope against exact classical Lindblad results and, where feasible, against a closed-system autocorrelation calculation.

4.7 Reservoir parameterization and current conventions

The boundary-only model above also provides a direct comparison between relaxation to a spatially uniform state and relaxation to a nonequilibrium steady state with a nonuniform population profile. Write the reservoirs as

$$V_{b,+} = \sqrt{\Gamma_b p_b}\,\sigma^+_{s_b}, \qquad V_{b,-} = \sqrt{\Gamma_b(1-p_b)}\,\sigma^-_{s_b}, \qquad s_L = 1, \quad s_R = n, \tag{55}$$

where $0 < p_b < 1$ and $\Gamma_b > 0$. The number p_b is the preferred excitation population of reservoir b, not its preferred Z magnetization. For an isolated boundary spin,

$$n_j = \frac{\mathbb{I} - Z_j}{2}, \qquad \frac{d}{dt}\langle n_{s_b}\rangle = \Gamma_b\big(p_b - \langle n_{s_b}\rangle\big), \qquad \langle Z_{s_b}\rangle_* = 1 - 2p_b. \tag{56}$$

In particular, the original boundary jumps correspond to $p_L = (1+\mu)/2$ and $p_R = (1-\mu)/2$; in our computational convention the left reservoir favors $Z = -\mu$ and the right favors $Z = +\mu$.

Define the positive left-to-right *excitation* current by

$$I_\ell = \frac{J}{2}\,(Y_\ell X_{\ell+1} - X_\ell Y_{\ell+1}) = -\frac{1}{2} j_\ell. \tag{57}$$

The continuity equations are

$$\frac{d}{dt}\langle n_1\rangle = \Gamma_L(p_L - \langle n_1\rangle) - \langle I_1\rangle, \tag{58}$$

$$\frac{d}{dt}\langle n_j\rangle = \langle I_{j-1}\rangle - \langle I_j\rangle, \qquad 2 \le j \le n-1, \tag{59}$$

$$\frac{d}{dt}\langle n_n\rangle = \langle I_{n-1}\rangle + \Gamma_R(p_R - \langle n_n\rangle). \tag{60}$$

Thus a stationary population profile can be nonuniform, while the stationary current must be equal on all bonds. The current j_ℓ defined earlier is instead the Z-magnetization current.

4.8 Experiment A: uniform stationary populations from equal reservoirs

Choose $p_L = p_R = p$. For the XXZ Hamiltonian, an exact stationary state is

$$\rho_* = \tau_p^{\otimes n}, \qquad \tau_p = (1-p)\,|0\rangle\langle 0| + p\,|1\rangle\langle 1|\,. \tag{61}$$

Indeed, each boundary dissipator annihilates this state. Moreover,

$$\rho_* \propto e^{-\alpha N}, \qquad N = \sum_{j=1}^{n} n_j, \qquad \alpha = \log\frac{1-p}{p}, \qquad [H_{\mathrm{XXZ}}, N] = 0, \tag{62}$$

so its Hamiltonian commutator vanishes. For a connected finite chain with $J \neq 0$ and positive pumping and loss rates at the boundaries, this state is unique and attracting. One way to see the absence of a protected sector is that commuting with both raising and lowering operators at site 1 forces an operator to be identity there; commuting also with the exchange Hamiltonian propagates this constraint through the connected chain. Consequently,

$$\lim_{t\to\infty}\langle Z_j(t)\rangle = 1 - 2p \quad \text{for every } j, \qquad \lim_{t\to\infty}\langle I_\ell(t)\rangle = 0. \tag{63}$$

This is a reservoir-selected equilibrium state; it is generally not the finite-temperature Gibbs state of the interacting Hamiltonian.

For a concrete three-qubit benchmark, take

$$n = 3, \qquad J = 1, \qquad \Delta = 0, \qquad \Gamma_L = \Gamma_R = 2, \qquad p_L = p_R = \frac{1}{4}. \tag{64}$$

Then the exact target is

$$(\langle Z_1\rangle_*, \langle Z_2\rangle_*, \langle Z_3\rangle_*) = (0.5, 0.5, 0.5). \tag{65}$$

Prepare $|000\rangle$ and $|111\rangle$ in separate experiments to demonstrate convergence from opposite initial polarizations. A third initial state, $|100\rangle$, demonstrates relaxation from a spatially nonuniform population.

Essential nonzero equilibrium target and noise controls. The choice $p = 1/4$, rather than $p = 1/2$, is deliberate: every site must converge to the nonzero magnetization $\langle Z_j\rangle_* = 0.5$. Simple depolarizing noise instead drives $\langle Z_j\rangle$ toward zero, while uncontrolled amplitude damping favors $\langle Z_j\rangle = +1$. Thus the chosen target lies away from both common noise limits. A nonzero plateau alone does not separate all circuit errors: pumping, relaxation, and readout bias can combine to produce an incorrect nonzero plateau. Require agreement with the predicted value from both initial states, and compare to the noiseless compiled circuit. As an optional reservoir-control run, change both reservoirs to $p = 3/4$; the exact uniform target becomes $\langle Z_j\rangle_* = -0.5$ without changing the Hamiltonian or total boundary rates. Observing both signs under this controlled change is stronger evidence for reservoir-selected equilibration than observing one plateau alone.

4.9 Experiment B: nonuniform stationary populations from unequal reservoirs

Keep the Hamiltonian, chain length, and total boundary rates in eq. (64), and change only

$$p_L = 0.9, \qquad p_R = 0.1. \tag{66}$$

The four jump operators are now

$$V_{L,+} = \sqrt{1.8}\,\sigma_1^+, \qquad V_{L,-} = \sqrt{0.2}\,\sigma_1^-, \tag{67}$$

$$V_{R,+} = \sqrt{0.2}\,\sigma_3^+, \qquad V_{R,-} = \sqrt{1.8}\,\sigma_3^-. \tag{68}$$

All dissipation remains confined to the two endpoints. For a three-site XX chain with equal rates $\Gamma_L = \Gamma_R = \Gamma$, the exact stationary current and populations are

$$I_* = \frac{2J^2\Gamma}{\Gamma^2 + 4J^2}(p_L - p_R), \tag{69}$$

$$\langle n_1\rangle_* = p_L - \frac{I_*}{\Gamma}, \qquad \langle n_2\rangle_* = \frac{p_L + p_R}{2}, \qquad \langle n_3\rangle_* = p_R + \frac{I_*}{\Gamma}. \tag{70}$$

These identities follow from the stationary equations for the one-body correlations of the XX chain, together with boundary continuity. With the chosen parameters,

$$(\langle n_1\rangle_*, \langle n_2\rangle_*, \langle n_3\rangle_*) = (0.7, 0.5, 0.3), \qquad (\langle Z_1\rangle_*, \langle Z_2\rangle_*, \langle Z_3\rangle_*) = (-0.4, 0, 0.4), \tag{71}$$

$$\langle I_1\rangle_* = \langle I_2\rangle_* = 0.4, \qquad \langle j_1\rangle_* = \langle j_2\rangle_* = -0.8. \tag{72}$$

This is a nonequilibrium steady state: local populations are time-independent, but a persistent excitation current flows from left to right. Swapping the reservoir populations reverses the profile and current, providing a useful experimental control. The exact targets, also checked by solving the full three-qubit stationary Lindblad equation, are

(p_L, p_R)	$(\langle Z_1\rangle_*, \langle Z_2\rangle_*, \langle Z_3\rangle_*)$	$(\langle I_1\rangle_*, \langle I_2\rangle_*)$
$(0.25, 0.25)$	$(0.5, 0.5, 0.5)$	$(0, 0)$
$(0.9, 0.1)$	$(-0.4, 0, 0.4)$	$(0.4, 0.4)$
$(0.1, 0.9)$	$(0.4, 0, -0.4)$	$(-0.4, -0.4)$

Limitation for longer chains. The homogeneous boundary-driven XX chain has a flat interior stationary population profile, with deviations at the endpoints. The three-site example therefore has three distinct site values, but increasing n does not by itself produce a gradual bulk gradient. For a longer chain with a resolved bulk gradient while retaining boundary-only jumps, investigate an interacting XXZ regime and verify its finite-size profile classically before hardware execution. Adding bulk dephasing is another route to diffusive profiles, but changes the boundary-only model.

4.10 Measurement and convergence protocol on IBM Phoenix

Use the existing repeated-dilation implementation, with unconditional ancilla reset and no postselection. The backend identifier is `ibm_phoenix`; IBM documents mid-circuit reset and dynamic-circuit support for this device.[1] The measurement protocol is:

1. Measure all three system qubits in the Z basis to estimate the entire magnetization profile in one setting. Plot each $\langle Z_j(t)\rangle$ against time, and plot the late-time values against site index.

2. To measure both bond currents, add the settings XYX and YXY. Together they supply X_1Y_2, Y_1X_2, X_2Y_3, and Y_2X_3.

3. Start with dimensionless times $t \in \{0, 0.5, 1, 2, 3, 4, 6\}$, with $J = 1$. Exact continuous-time Lindblad integration gives magnetizations within 0.005 of their stationary targets at $t = 6$ for both experiments, starting from either $|000\rangle$ or $|111\rangle$. These are ideal-model predictions, not a guarantee of hardware accuracy.

4. Require both a late-time plateau and agreement between the final profiles from different initial states. In Experiment B, also check that the two currents agree and satisfy the boundary balance equations.

5. Use 4096 shots per time and basis setting as an initial statistical budget. The standard error of a single Z_j estimate is at most $1/\sqrt{4096} = 0.015625$ before mitigation and systematic errors, compared with the 0.4 separation between adjacent stationary magnetizations in Experiment B.

6. Compare hardware data with both exact Lindblad evolution and the actual noiseless discretized circuits. A finite step size can shift the stationary state. Choose the step size using the students' compiled circuit depth and noiseless convergence study; do not interpret every observed plateau as the intended Lindblad steady state.

For numerical diagnostics define, with the target appropriate to each run,

$$E_Z(t) = \max_{1\le j\le 3} |\langle Z_j(t)\rangle - \langle Z_j\rangle_*|\,, \qquad E_I(t) = |\langle I_1(t)\rangle - \langle I_2(t)\rangle|\,. \tag{73}$$

Report these with statistical uncertainties; E_I alone does not certify the correct steady state, since two equally biased current estimates can agree.

[1] IBM Quantum Compute Service changelog, September 2026: `https://quantum.cloud.ibm.com/docs/en/guides/changelog-quantum-compute-service`.

5 Model problem III: quantum-detailed-balance thermalization

5.1 Target Gibbs state and QDB condition

For thermalization, choose a model whose target Gibbs state is known:

$$\rho_\beta = \frac{e^{-\beta H_{\rm th}}}{Z_\beta}, \qquad Z_\beta = \mathrm{Tr}(e^{-\beta H_{\rm th}}). \tag{74}$$

A Lindblad generator satisfies quantum detailed balance (QDB) with respect to ρ_β if it is self-adjoint under the appropriate ρ_β-weighted inner product, such as the GNS/KMS inner product. A simple hardware-friendly construction is a diagonal, reversible, single-spin-flip dynamics. This is a classical detailed-balance Markov chain embedded into a Lindblad equation, and it is the right first experiment before attempting noncommuting Gibbs samplers.

Take the two-qubit Ising Hamiltonian

$$H_{\rm th} = -h(Z_1 + Z_2) - KZ_1Z_2. \tag{75}$$

For $z_1, z_2 \in \{+1, -1\}$, define

$$E(z_1, z_2) = -h(z_1 + z_2) - Kz_1z_2. \tag{76}$$

The target Gibbs probabilities are

$$\pi_\beta(z_1, z_2) = \frac{e^{-\beta E(z_1,z_2)}}{\sum_{z_1,z_2} e^{-\beta E(z_1,z_2)}}. \tag{77}$$

5.2 Explicit QDB jump operators

Let

$$P_s^{(q)} = \frac{\mathbb{I} + sZ_q}{2}, \qquad s \in \{+1, -1\}. \tag{78}$$

Conditioned on the neighboring spin value s, flipping qubit q from $|1\rangle$ to $|0\rangle$ or from $|0\rangle$ to $|1\rangle$ changes the Ising energy by a known amount. With our convention $Z|0\rangle = +|0\rangle$ and $Z|1\rangle = -|1\rangle$, define bounded detailed-balance rates

$$r_s^+ = \eta \frac{e^{\beta(h+Ks)}}{2\cosh[\beta(h+Ks)]}, \qquad r_s^- = \eta \frac{e^{-\beta(h+Ks)}}{2\cosh[\beta(h+Ks)]}. \tag{79}$$

Then

$$\frac{r_s^+}{r_s^-} = e^{2\beta(h+Ks)}, \tag{80}$$

which is exactly the Boltzmann ratio for the two directions of the conditional spin flip.

Use eight conditional jump operators:

$$V_{1,s}^+ = \sqrt{r_s^+}\,\sigma_1^+ P_s^{(2)}, \qquad V_{1,s}^- = \sqrt{r_s^-}\,\sigma_1^- P_s^{(2)}, \tag{81}$$

$$V_{2,s}^+ = \sqrt{r_s^+}\,P_s^{(1)}\sigma_2^+, \qquad V_{2,s}^- = \sqrt{r_s^-}\,P_s^{(1)}\sigma_2^-, \qquad s = \pm 1. \tag{82}$$

For diagonal density matrices, this is exactly a reversible continuous-time Markov chain with stationary distribution π_β. Since each forward/backward transition pair satisfies eq. (80), the generator

obeys detailed balance. The corresponding embedded Lindblad dynamics has ρ_β as a stationary state. If coherent off-diagonal components are present in ρ_0, add optional dephasing jumps

$$V_{\phi,q} = \sqrt{\gamma_\phi}\, Z_q, \tag{83}$$

which commute with H_{th} and leave ρ_β invariant.

Recommended parameters:

$$h = 0.3, \qquad K = 0.7, \qquad \beta \in \{0.5, 1, 2\}, \qquad \eta \in \{0.1, 0.2, 0.5\}. \tag{84}$$

The case $\beta = 0$ should be included as a sanity check: the target is the maximally mixed state and all conditional forward/backward rates are equal.

5.3 Dilated Hamiltonian for the thermalization generator

Ignoring optional dephasing, there are $J_{\text{jump}} = 8$ jumps. The active ancilla dimension is 9, so use four ancilla qubits. Label the eight active nonzero ancilla states by

$$\alpha \in \{(1,+,+), (1,+,-), (1,-,+), (1,-,-), (2,+,+), (2,+,-), (2,-,+), (2,-,-)\}, \tag{85}$$

where the entries denote site, flip direction, and neighbor condition. Let V_α denote the corresponding operator in eqs. (81) and (82). The first-order dilated Hamiltonian is

$$\widetilde{H}^{\text{th}}_{\Delta t} = \sqrt{\Delta t}\; |0\rangle\langle 0|_a \otimes H_{\text{th}} + \sum_{\alpha=1}^{8} \left(|\alpha\rangle\, \langle 0|_a \otimes V_\alpha + |0\rangle\, \langle \alpha|_a \otimes V_\alpha^\dagger \right). \tag{86}$$

The unitary step is

$$U^{\text{th}}_{\Delta t} = \exp\!\left(-i\sqrt{\Delta t}\, \widetilde{H}^{\text{th}}_{\Delta t}\right). \tag{87}$$

If dephasing jumps are included, append them to the same sum, increasing the active ancilla dimension accordingly.

5.4 Trotter simplifications for QDB jumps

The Hamiltonian H_{th} is diagonal, so $e^{-i\Delta t|0\rangle\langle 0|_a \otimes H_{\text{th}}}$ consists only of controlled Z_1, Z_2, and $Z_1 Z_2$ rotations. Each conditional jump is a projected amplitude-raising or amplitude-lowering exchange. For example,

$$|\alpha\rangle\, \langle 0|_a \otimes \sqrt{r_s^+}\, \sigma_1^+ P_s^{(2)} + \text{h.c.} \tag{88}$$

acts only when the neighbor qubit is in the $Z = s$ sector. Thus it can be compiled as a controlled exchange between the ancilla transition $|0\rangle_a \leftrightarrow |\alpha\rangle_a$ and the system qubit, with an additional control on $P_s^{(2)}$.

Useful optimizations:

- jumps with orthogonal projectors $P_+^{(q)}$ and $P_-^{(q)}$ can be scheduled in either order because they act on disjoint system subspaces;
- the two Pauli strings inside each σ^+ or σ^- exchange commute as in the amplitude-damping example;
- if hardware resources are tight, first run a one-qubit Gibbs sampler with two jumps $\sqrt{r^+}\sigma^+$ and $\sqrt{r^-}\sigma^-$, then the two-qubit conditional sampler.

5.5 Certification of thermal convergence

Because H_{th} is diagonal, certification can avoid full tomography. Measure computational-basis populations

$$p_t(z_1, z_2) = \Pr_t(Z_1 = z_1, Z_2 = z_2) \tag{89}$$

and compare against $\pi_\beta(z_1, z_2)$. Recommended metrics are

$$d_{\mathrm{TV}}(t) = \frac{1}{2} \sum_{z_1,z_2} |p_t(z_1, z_2) - \pi_\beta(z_1, z_2)|, \tag{90}$$

$$D_{\mathrm{KL}}(p_t \| \pi_\beta) = \sum_{z_1,z_2} p_t(z_1, z_2) \log \frac{p_t(z_1, z_2)}{\pi_\beta(z_1, z_2)}, \tag{91}$$

$$E(t) = \sum_{z_1,z_2} p_t(z_1, z_2) E(z_1, z_2), \tag{92}$$

$$C_{12}(t) = \langle Z_1 Z_2 \rangle_t. \tag{93}$$

For a two-qubit experiment, also perform occasional full state tomography to check that coherences are small. A practical convergence fit is

$$|E(t) - E_\beta| \approx A e^{-\lambda t}, \tag{94}$$

where λ is an empirical mixing rate. Choose $T \gtrsim 3/\lambda$ for a first plateau and $T \gtrsim 5/\lambda$ for quantitative comparison.

The thermalization experiment connects to recent work on dissipative Gibbs-state preparation and quantum detailed balance [17, 2, 1, 18]. The proposed two-qubit model is deliberately simpler than those many-body Gibbs samplers, but it tests exactly the hardware primitive needed by those algorithms: repeated, unconditional simulation of a detailed-balanced Lindbladian.

6 Benchmarking and error-mitigation plan

6.1 What has already been benchmarked

Related hardware demonstrations include QITE-based simulation of Lindblad dynamics on IBM hardware [10], adaptive variational Lindblad simulation on IBM quantum processors [3], and mid-circuit/dynamic-circuit benchmarking and qubit reuse [7, 19, 9, 5]. These works motivate the present project but do not replace it: the specific benchmark here is the first-order Hamiltonian dilation with repeated ancilla reset, explicit $\sqrt{\Delta t}$ dissipative rotations, and physical observables such as steady current and Gibbs convergence.

6.2 Three error sources to separate

For each model, report at least three comparisons:

1. exact Lindblad integration versus exact first-order dilation channel;
2. exact first-order dilation channel versus Trotterized/synthesized dilation unitary on a noiseless simulator;
3. noisy hardware result versus noiseless Trotterized circuit.

This separation is crucial. Otherwise one cannot tell whether failure comes from Lindblad time discretization, Hamiltonian-simulation/Trotter error, reset/readout error, or ordinary gate noise.

6.3 Step-size extrapolation and ZNE

For a scalar observable O, collect estimates

$$f(\Delta t) = \mathrm{Tr}[O\Phi_{\Delta t}^{T/\Delta t}(\rho_0)]. \tag{95}$$

For the first-order method,

$$f(\Delta t) = f(0) + c_1\Delta t + c_2\Delta t^2 + \cdots, \tag{96}$$

until hardware noise dominates. Use Richardson or least-squares extrapolation over several Δt values. Chebyshev-like step grids should be tested because they can reduce interpolation variance compared with equally spaced grids [14].

At each Δt, also consider zero-noise extrapolation by gate folding or calibrated noise scaling. The desired two-stage estimator is

$$f_{\mathrm{ideal}}(0) \approx \lim_{\Delta t\to 0}\lim_{\lambda_{\mathrm{noise}}\to 0} f_{\mathrm{hardware}}(\Delta t, \lambda_{\mathrm{noise}}). \tag{97}$$

A practical question is whether the order of these extrapolations matters within statistical uncertainty.

7 Concrete summer milestones

Weeks 1–2: software reference. Implement exact Lindblad integration, exact dilation channel, and Trotterized dilation circuits for the two-qubit model. Produce plots of $n_1(t)$, $n_2(t)$, $C_{XY}(t)$, and $j_{12}(t)$.

Weeks 3–4: first IBM runs. Run the two-qubit damping model with both fresh ancillas and mid-circuit reset. Quantify reset/readout overhead by comparing them at the same effective T and Δt.

Weeks 5–7: step-size extrapolation and ZNE. Run $\Delta t \in \{0.2, 0.1, 0.05\}$ or a Chebyshev-like grid. Compare raw, step-extrapolated, ZNE-only, and combined estimators.

Weeks 8–10: boundary-driven current. Run the $n = 2$ boundary-driven model first, then $n = 3$ if depth allows. Target the sign, magnitude, and μ-linearity of $\langle j\rangle$.

Weeks 11–12: QDB thermalization. Run the one-qubit Gibbs sampler as a calibration, then the two-qubit conditional-jump sampler. Certify convergence by total-variation distance to the exact Gibbs distribution and by energy relaxation.

8 Implementation checklist

- Fix the Pauli and raising/lowering convention before coding.
- Build $\widetilde{H}_{\Delta t}$ symbolically and verify that $\Phi_{\Delta t}(\rho) = \rho + \Delta t\mathcal{L}(\rho) + O(\Delta t^2)$ numerically.
- Validate every compiled $U_{\Delta t}$ against exact matrix exponentiation for one step.

- Always record ancilla outcomes, but do not postselect on them.
- Use the same transpilation and layout strategy across all Δt values in an extrapolation study.
- For every plotted hardware observable, include exact Lindblad, exact dilation, noiseless Trotter, and hardware data when feasible.
- Report shot counts and uncertainty bars; extrapolation without error bars is not meaningful.

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